

5ME42A(ME) – Manufacturing Technology

5th Semester B.E. Mechanical Engineering

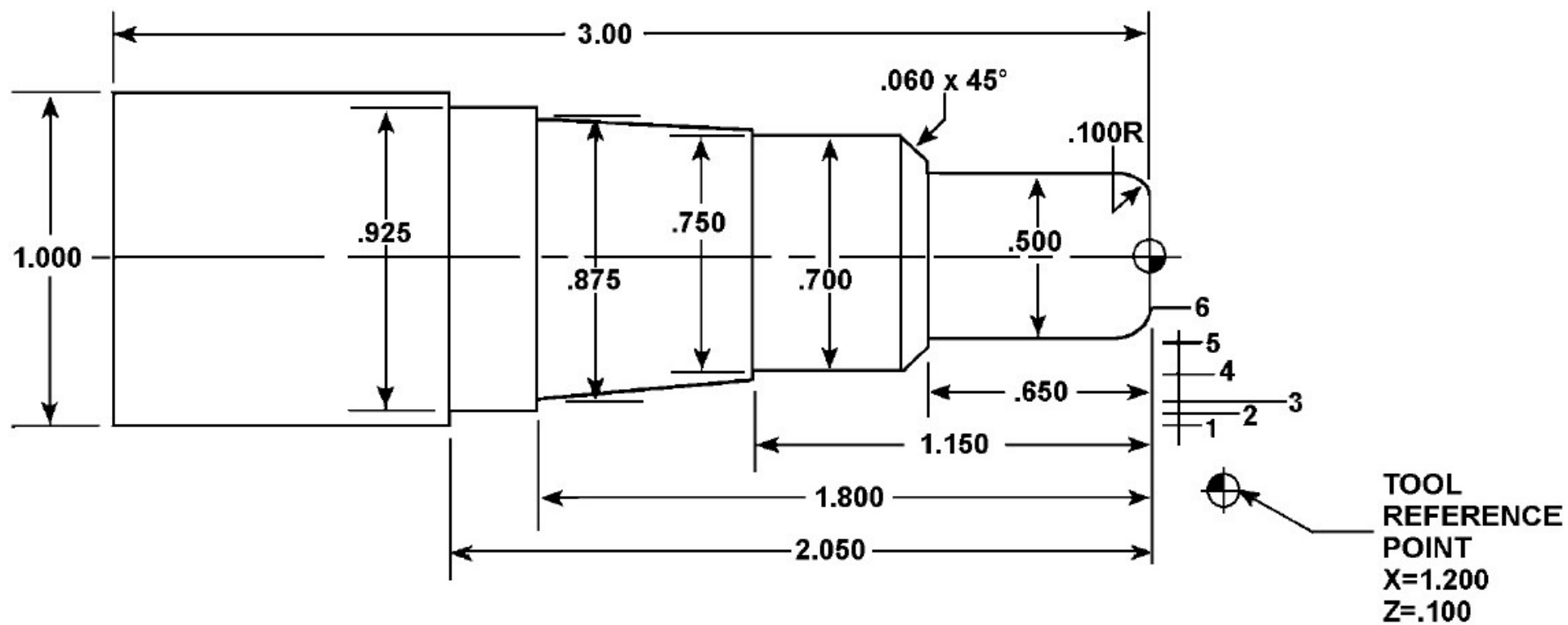
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Turning Programming

Programming Sequence

- N05 G20 G90

G20 inch data input, G90 absolute positioning mode

- N10 G95 G96 S2000 M03

G95 feed rate per revolution, G96 constant feed rate, S2000 spindle speed set at 2000 r/min, M03 spindle ON clockwise

- N15 T0202

tool number

- N20 G00 X1.200 Z.100

G00 rapid traverse mode, X&Z tool reference or change point, X1.200 tool point .100 away from the outside diameter, Z.100 tool point .100 to the right of end of work

- N25 G00 X.300 Z.050
G00 rapid traverse mode, X.300 tool point at .300 diameter for start of .100 radius, Z.050 tool point .050 away from end of the part
- N30 G01 Z0
G01 linear interpolation (feed), Z0 tool point touching end of the work
- N35 G03 X.500 Z-.100 R.100
G03 circular interpolation (counterclockwise), X.500 largest diameter of radius, Z-.100 end of radius on .500 diameter, R.100 size of the radius
- N40 G01 Z-.650
G01 linear interpolation, Z-.650 machines .500 diameter to .650 length
- N45 X.580
X.580 tool moves out to the small diameter of .060 x 450 bevel
- N50 X.700 Z-.710
X.700 large diameter of bevel, Z-.710 end distance of bevel
- N55 Z-1.150
Z-1.150 the .700 diameter cut to 1.150 length
- N60 X.750
X.750 cutting tool feeds out to .750 (small end of taper)
- N65 X.875 Z-1.800 (cutting taper)
X.875 large end of taper, Z-1.800 length that taper is cut.

- N70 X.925

X.925 tool feeds out (faces) to .925 diameter

- N75 Z-2.050

Z-2.050 the .925 diameter is cut to 2.050 length

- N80 X1.050

X1.050 the tool is fed out to .050 past the diameter of the part

- N85 G00 X1.200 Z.100 (tool back to tool reference point)

G00 rapid traverse mode, X1.200 & Z.100 (reference point positions)

- N90 M30

M30 end of program

Milling and Drilling Programming

1. Program in the absolute mode starting at the tool change position at the top left corner of the print.
2. The material is aluminum (300 CS), feedrate 10 in/min.
3. The cutting tool is a .250 in. diameter high speed steel 2-flute end mill.
4. Mill the 1 in. square slot.
5. Drill the two .250 in. diameter holes, .250 in. deep.
6. Mill the .250 in. wide angular slot, .125 in. deep.
7. Mill the .250 in. wide circular groove, .125 in. deep.
8. After the job is completed, return to the tool change position.

Milling and Drilling Programming

- N5 G92 X-1.000 Y1.000 Z1.000

G92 programmed offset of reference point (tool change position), X-1.000 tool set at 1.000 to the left of the part, Y1.000 tool set at 1.000 above the top edge of the part, Z1.000 the end of the cutter is 1.000 above the top surface of the part

- N10 G20 G90

G20 inch data input, G90 absolute programming mode

- N15 M06 T01

M06 tool change command, T01 tool no. 1 (.250 diameter, 2-flute end mill)

- N20 S2000 M03

S2000 spindle speed set at 2000 r/min., M03 spindle on clockwise

- N25 G00 X0 Y0 Z.100

G00 rapid traverse rate to X0 Y0 at the top left corner of the part, Z.100 tool rapids down to within .100 of the work surface.

Machining the square groove

- N30 X.375 Y-.375

tool rapids to position A

- N35 G01 Z-.125 F10

G01 linear interpolation, Z-.125 tool feeds .125 below the work surface, F10 feed rate set at 10 in./min.

- N40 X1.625 Y-.375
X1.625 top groove cut to the right hand end, Y-.375 measurement did not change because it was set in block N30
- N45 Y-1.625
Y-1.625 right hand side of the groove cut
- N50 X.375
X.375 bottom groove cut to the left side
- N55 Y-.375
Y-.375 left-hand side of groove cut; this completes the groove
- N60 G00 Z.100
G00 rapid traverse mode, Z.100 tool rapids to .100 above work surface

Hole Drilling

- N65 G00 X.875 Y-.750
tool rapids to the top left hole location
- N70 G01 Z-.250 F10
tool feeds .250 into work at 10 in./min. to drill the first hole
- N75 G00 Z.100
tool rapids out of hole to .100 above work surface
- N80 X1.250 Y-1.125
tool rapids to second hole location

- N85 G01 Z-.250 F10
tool feeds .250 into work at 10 in./min. to drill the second hole
- N90 G00 Z.100
tool rapids out of hole to .100 above work surface

Machining the Angular Slot

- N95 X1.125 Y-.875 (location B)
tool rapids to the start of the angular slot
- N100 G01 Z-.125 F10
G01 linear interpolation, Z-.125 tool feeds to .125 below the work surface, F10 feed rate set at 10 in./min
- N105 X1.250 Y-.750
angular slot cut to top right corner
- N110 G00 Z.100
tool rapids to .100 above work surface

Machining the Circular Groove

- N115 X.750 Y-1.000 (location C)
tool rapids to start of circular groove
- N120 G01 Z-.125 F10
tool feeds to .125 below the work surface

- N125 G03 X1.000 Y-1.250 R.250
G03 circular interpolation counterclockwise, X & Y location of end of circular groove, R.250 radius of arc is .250
- N130 G00 Z.100
tool rapids to .100 above work surface
- N135 X-1.000 Y1.000
tool rapids back to tool change position
- N140 M05
M05 spindle turned off.
- N145 M30
M30 end of program